Lecture 4 Tutorial — Data Distributions

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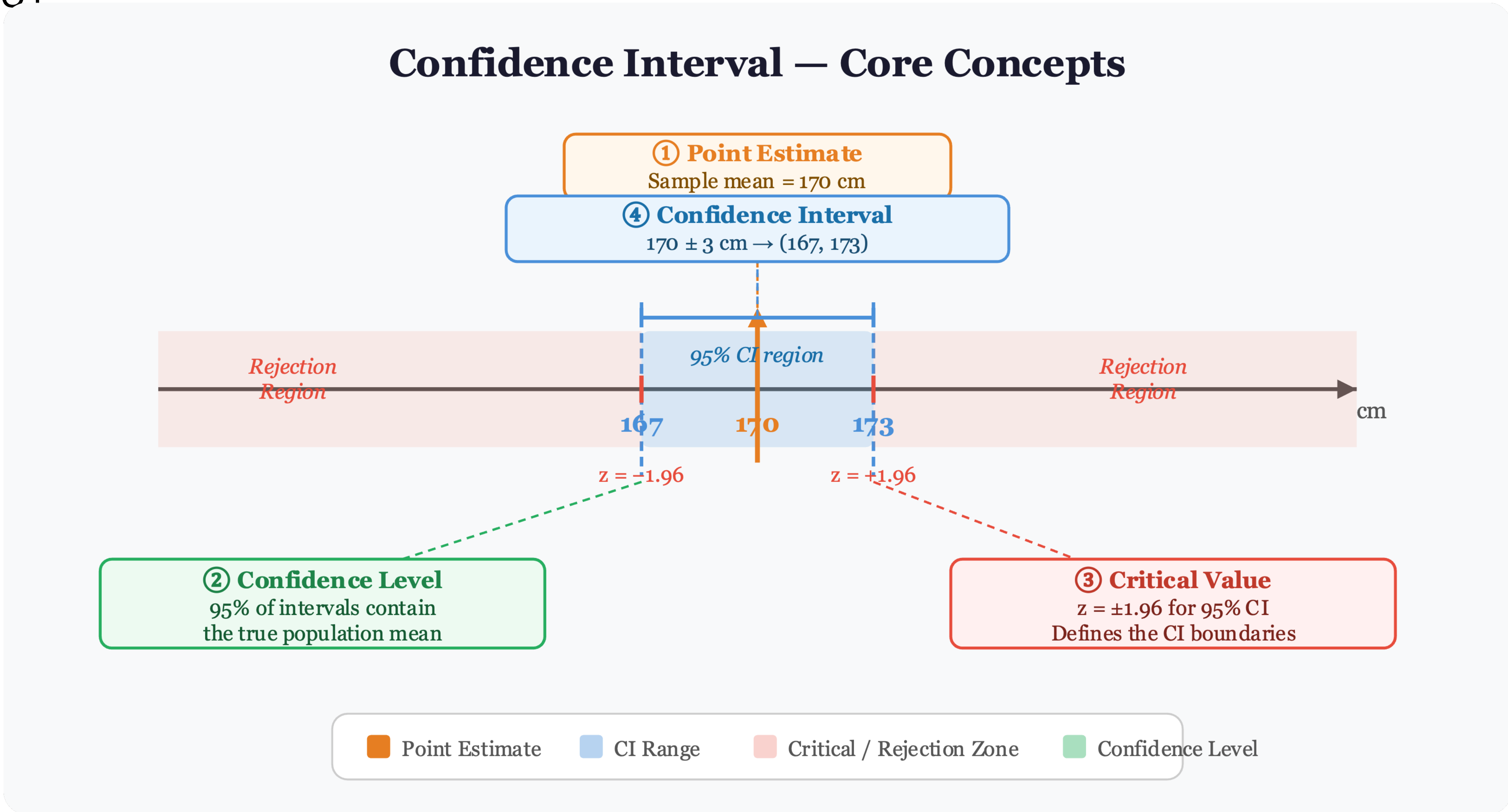
Q1 Complete the definitions.

1. A ______ estimate is a single value used to estimate a population parameter.
2. The ______ level is (for example, 0.95).
3. For the standard normal distribution, a ______ value is a z score on the borderline separating significantly low/high values.
4. A ______ interval gives a range of plausible values for a population parameter.

01



Confidence Interval — Core Concepts
① Point Estimate
Sample mean = 170 cm
④ Confidence Interval
170 ± 3 cm → (167, 173)
95% CI region
Rejection Region
Rejection Region
cm
167
170
173
z = −1.96
z = +1.96
② Confidence Level
95% of intervals contain
the true population mean
③ Critical Value
z = ±1.96 for 95% CI
Defines the CI boundaries
Point Estimate
CI Range
Critical / Rejection Zone
Confidence Level

Q1 - Solution

Question

1. A ______ estimate is a single value used to estimate a population parameter.
2. The ______ level is (for example, 0.95).
3. For the standard normal distribution, a ______ value is a z score on the borderline separating significantly low/high values.
4. A ______ interval gives a range of plausible values for a population parameter.

Answers

1. point
2. confidence
3. critical
4. confidence

Q2. Confidence Interval Interpretation (Multiple Choice)

Question

Choose the best interpretation of a 95% confidence interval for a population proportion p.

A. There is a 95% chance that the true value of p is in this specific interval.

B. We are 95% confident that the interval produced by this method contains the true value of p.

C. 95% of sample proportions must fall inside this interval.

D. The population proportion changes with probability 0.95.

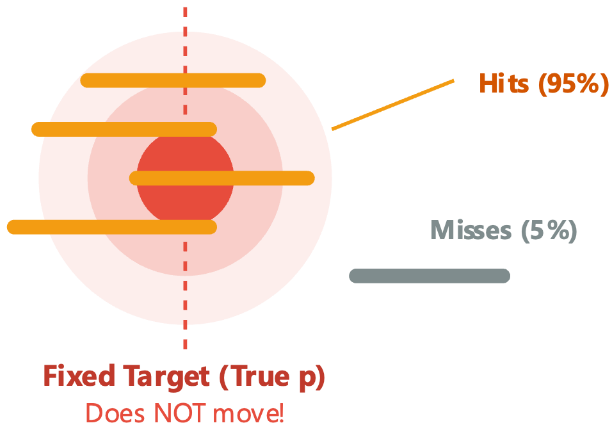
Q2 — Interpretation of a 95% Confidence Interval

Why is 95% about the "Method", not the "Parameter"?

✓ CORRECT ANSWER: OPTION B

"We are 95% confident in the METHOD."

The true population proportion (p) is a fixed target.
Our calculation method is like shooting arrows.



Summary of B:
If we repeat this process 100 times,
about 95 of the intervals will capture p.

OPTION A: "95% chance p is in the interval"

Why it's wrong:
Once a specific interval is calculated, the true p
is either in it (100%) or it isn't (0%).

Probability applies to the method, not a fixed constant.

OPTION C: "95% of sample proportions..."

Why it's wrong:
A confidence interval is built to estimate the
population parameter (p), not to predict where
future sample statistics ($\hat{p}$) will fall.

OPTION D: "Population proportion changes..."

Why it's wrong:
The population proportion p is an objective reality
in the world. The target does not jump around!

It is a fixed constant, it never moves randomly.

PARAMETER
Fixed (Population Truth)

INTERVAL
Random (Varies by Sample)

CONFIDENCE
In the Method (Success Rate)

Q3. Confidence Level and Tails (Calculation Fill-in)

Question

For a 95% confidence level, complete the values:

1. $\alpha =$
2. $\alpha / 2 =$
3. $1 - \alpha / 2 =$

The cumulative area to the left of the positive critical z value is ______.

Q3 — Confidence Level, Tails & Cumulative Area

Visualizing α, α/2, and the cumulative area to the left of the positive critical z-value

Step-by-Step Calculation

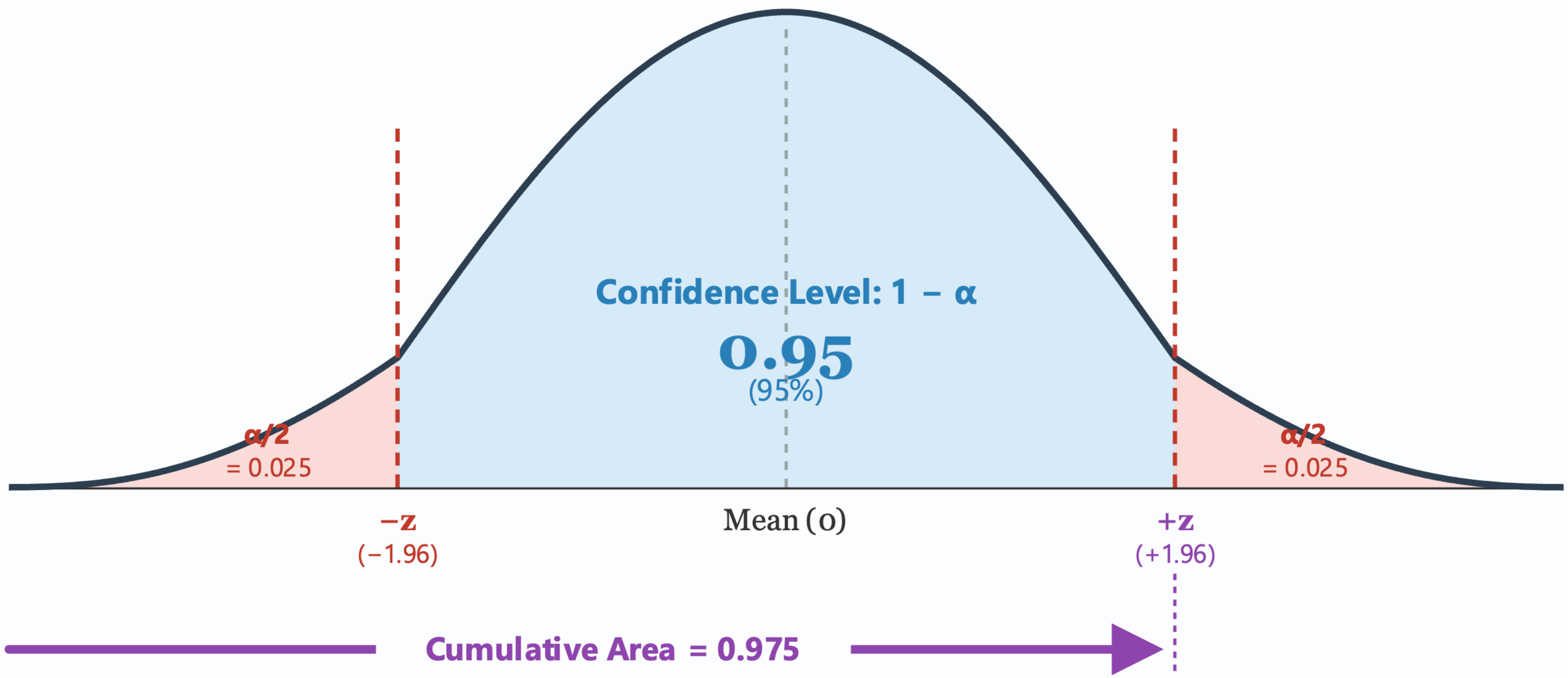
1. α (Total Error Rate)	$= 1 - 0.95 = \mathbf{0.05}$ (Both tails combined)
2. α / 2 (Single Tail)	$= 0.05 \div 2 = \mathbf{0.025}$ (Split equally)
3. Cumulative Area	$= 1 - 0.025 = \mathbf{0.975}$

Why Cumulative Area is 0.975?

Standard Z-tables show the **cumulative area to the LEFT**.
To find z for the right boundary, we must add up:

Left Tail (0.025) + Center (0.95) = 0.975

→ *Looking up 0.975 in the Z-table yields z = 1.96*



Q3. Confidence Level and Tails (Calculation Fill-in)

Question

For a 95% confidence level, complete the values:

1. α =
2. $\alpha / 2$ =
3. $1 - \alpha / 2$ =

The cumulative area to the left of the positive critical z value is ______.

Answers

1. $\alpha = 0.05$
2. $\alpha / 2 = 0.025$
3. $1 - \alpha / 2 = 0.975$

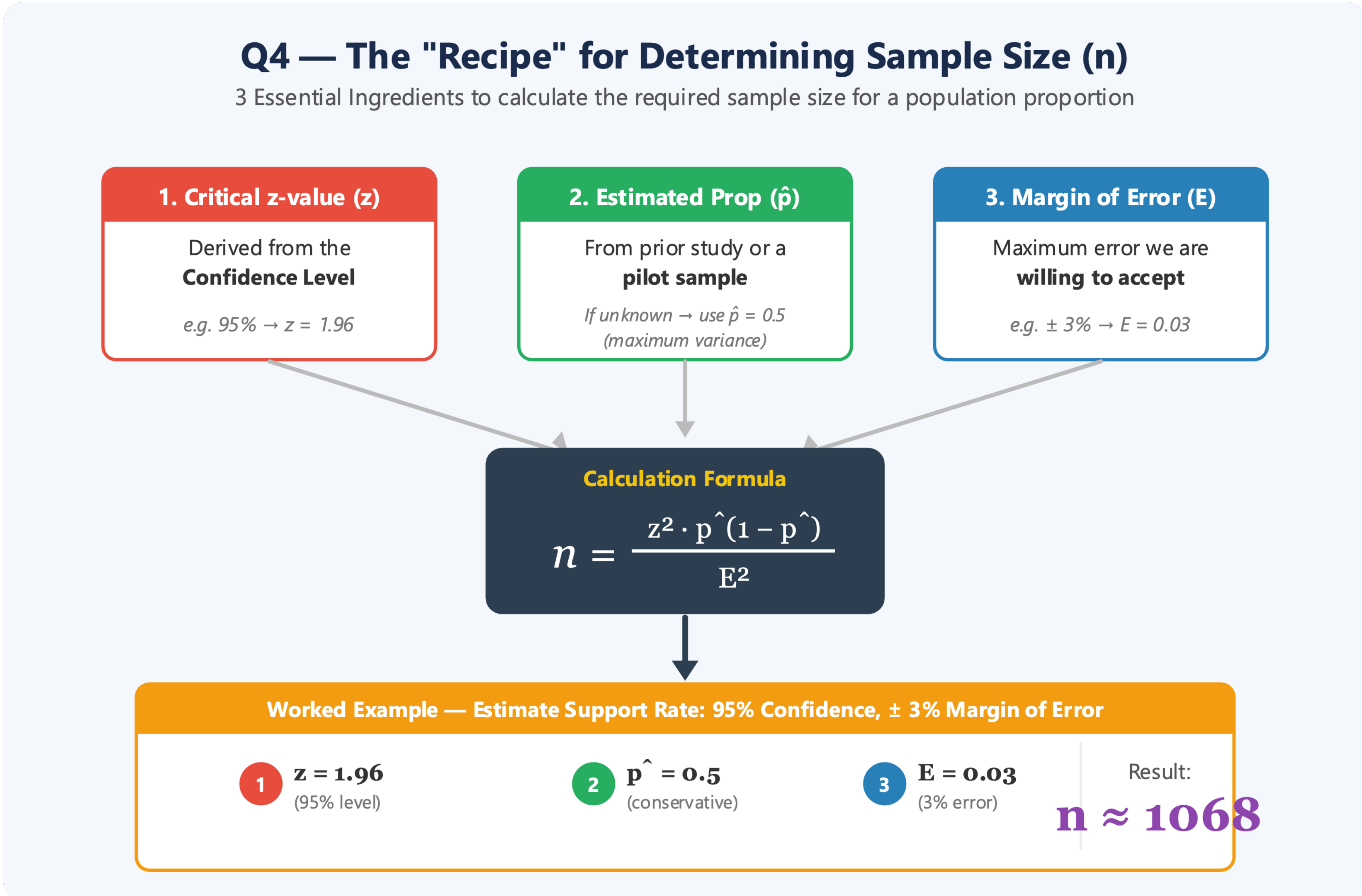
Cumulative area = 0.975

Q4. Determining Sample Size (Short Answer)

Question

List the three ingredients for sample size calculation for estimating population proportion.

Q4


Q4 — The "Recipe" for Determining Sample Size (n)
3 Essential Ingredients to calculate the required sample size for a population proportion
1. Critical z-value (z)
Derived from the **Confidence Level**
e.g. 95% → z = 1.96
2. Estimated Prop ($\hat{p}$)
From prior study or a **pilot sample**
If unknown → use $\hat{p}$ = 0.5 (maximum variance)
3. Margin of Error (E)
Maximum error we are **willing to accept**
e.g. ± 3% → E = 0.03
Calculation Formula
$n = \frac{z^2 \cdot \hat{p}(1-\hat{p})}{E^2}$
Worked Example — Estimate Support Rate: 95% Confidence, ± 3% Margin of Error
1
$z = 1.96$
(95% level)
2
$\hat{p} = 0.5$
(conservative)
3
$E = 0.03$
(3% error)
Result:
$n \approx 1068$

Q4. Determining Sample Size (Short Answer)

Question

List the three ingredients for sample size calculation for estimating population proportion.

Answer

1. Critical z-value
2. Sample proportion (or estimated proportion)
3. Desired margin of error

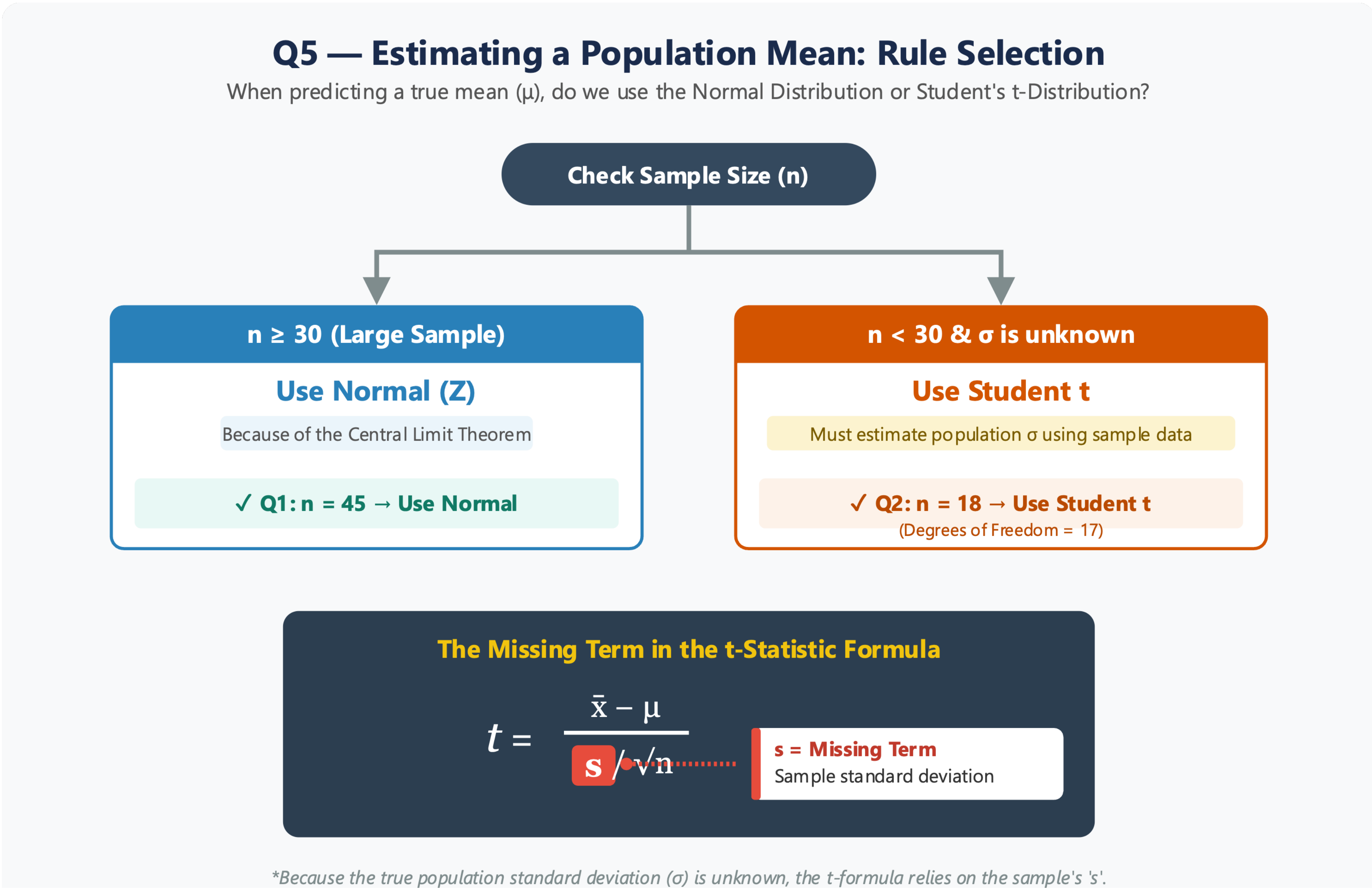
Q5. Estimating a Population Mean (Rule Selection)

Question

1. $n = 45$ → use (Normal / Student t)
2. $n = 18$ → use (Normal / Student t)

Complete: $t = (\bar{x} - \mu) / (\_\_\_\_ / \sqrt{n})$

Q5


Q5 — Estimating a Population Mean: Rule Selection
When predicting a true mean (μ), do we use the Normal Distribution or Student's t-Distribution?
Check Sample Size (n)
n ≥ 30 (Large Sample)
Use Normal (Z)
Because of the Central Limit Theorem
✓ Q1: n = 45 → Use Normal
n < 30 & σ is unknown
Use Student t
Must estimate population σ using sample data
✓ Q2: n = 18 → Use Student t
(Degrees of Freedom = 17)
The Missing Term in the t-Statistic Formula
t = (x̄ − μ) / (s / √n)
s = Missing Term
Sample standard deviation
*Because the true population standard deviation (σ) is unknown, the t-formula relies on the sample's 's'.

Q5 - Solution

Q5. Estimating a Population Mean (Rule Selection)

Question

1. $n = 45$ → use (Normal / Student t)
2. $n = 18$ → use (Normal / Student t)

Complete: $t = (\bar{x} - \mu) / (\_\_\_\_ / \sqrt{n})$

Answers

1. Normal
2. Student t

Missing term: s (sample standard deviation)

Q6. Estimating a Population Variance (Formula + Interpretation)

Question

Complete $\chi^2 = [(n-1)s^2] / $ ____

s^2 is (sample / population) variance.

σ^2 is (sample / population) variance.

Q6 — Estimating a Population Variance (χ^2)

Anatomy of the Chi-Square Test Statistic Formula

$\chi 2$ measures how much the **sample variance** s^2 differs from the **hypothesized population variance** σ^2.

Degrees of Freedom (df)

df = n − 1

Sample Variance (s^2)

Calculated from the sample data

$$\chi^2 = \frac{(n-1)\, s^2}{\boxed{\sigma^2}}$$

- $s^2 \approx \sigma^2 \to \chi^2 \approx n-1$
- $s^2 \gg \sigma^2 \to \chi^2$ is large

Q: Population Variance (σ^2)

This is the hypothesized value being tested. It goes in the DENOMINATOR.

Worked Example Calculation

Given Data:

- $\mathbf{n = 10}$ (so df = 10 − 1 = 9)
- $\mathbf{s^2 = 4}$ (sample variance)
- $\mathbf{\sigma^2 = 2}$ (hypothesized pop. variance)

Plug into formula:

$$\chi^2 = \frac{(9)(4)}{2} = \frac{36}{2} = \mathbf{18}$$

Q6 - Solution

Q6. Estimating a Population Variance (Formula + Interpretation)

Question

Complete $\chi^2 = [(n-1)s^2] / \_\_\_\_$

s^2 is (sample / population) variance.

σ^2 is (sample / population) variance.

Answers

Denominator: σ^2

s^2 = sample variance

σ^2 = population variance

Q7. Null and Alternative Hypotheses (Construction)

Scenario (from lecture theme): compare plant growth under Fertilizer A and Fertilizer B.

Construct hypotheses for the claim: "There is a difference in average plant growth between the two fertilizers."

$$H_0: \underline{\hspace{8cm}}$$

$$H_a: \underline{\hspace{8cm}}$$

Then identify the test type: ________________ (left-tailed / right-tailed / two-tailed)

Step 1. The Research Claim:

*"There is a **DIFFERENCE** in average plant growth between Fertilizer A and B."*

Null Hypothesis (H_0)

$$\mu 1 = \mu 2$$

Always states NO DIFFERENCE

(Equality / Status Quo)

Alternative Hypothesis (H_a)

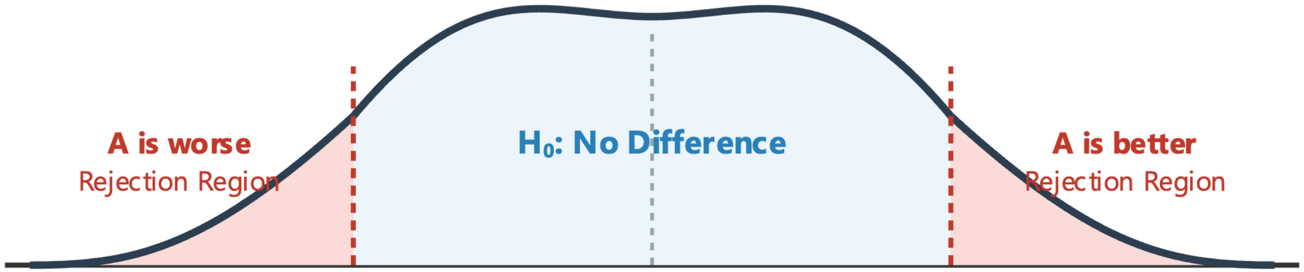
$$\mu 1 \neq \mu 2$$

Matches the Research Claim!

("Difference exists", can be greater or less)

Step 3. Test Type: TWO-TAILED TEST

Because "≠" means the difference could go in EITHER direction (A > B or A < B).



Q7. Null and Alternative Hypotheses (Construction)

Scenario (from lecture theme): compare plant growth under Fertilizer A and Fertilizer B.

Construct hypotheses for the claim: "There is a difference in average plant growth between the two fertilizers."

$$H_0 : \underline{\hspace{10cm}}$$

$$H_a : \underline{\hspace{10cm}}$$

Then identify the test type: ________________ (left-tailed / right-tailed / two-tailed)

Q7. Null and Alternative Hypotheses

$$H_0 : \ \mu_A = \mu_B$$

$$H_a : \ \mu_A \neq \mu_B$$

Test type: **two-tailed**

Q8. Significance Level and Error Interpretation (True / False + Correction)

Question

1. The significance level is the probability of rejecting H_0 when H_0 is true.
2. A larger α generally makes rejection of H_0 harder.
3. The parameter tested is always a mean.

TRUE **1. The significance level (α) is the probability of rejecting H_0 when H_0 is true.**

Explanation:

α represents the **Type I Error** — essentially a "False Alarm".

e.g., If $\alpha = 0.05$, there is a 5% risk of rejecting a true Null Hypothesis.

	H_0 is TRUE	H_0 is FALSE
Reject H_0	**Type I Error (α)**	Correct
Don't Reject		

FALSE ~~**2. A larger α generally makes rejection of H_0 harder.**~~ **→ It makes it EASIER.**

Why? Because a larger α physically **expands the rejection region**. A bigger target is easier to hit!

Small α (e.g., 0.01)
Tiny target → Hard to reject H_0

Large α (e.g., 0.10)
Big target → EASIER to reject H_0

FALSE ~~**3. The parameter tested is always a mean.**~~ **→ We can test ALL kinds of parameters!**

Hypothesis tests aren't exclusively for the population mean (μ). Depending on the real-world question, we test:

- μ **Mean** — e.g., Average plant growth
- p **Proportion** — e.g., Conversion rate / Polls
- σ^2 **Variance** — e.g., Robot sensor error

Q8. Significance Level and Error Interpretation (True / False + Correction)

Question

1. The significance level is the probability of rejecting H_0 when H_0 is true.
2. A larger α generally makes rejection of H_0 harder.
3. The parameter tested is always a mean.

Answers

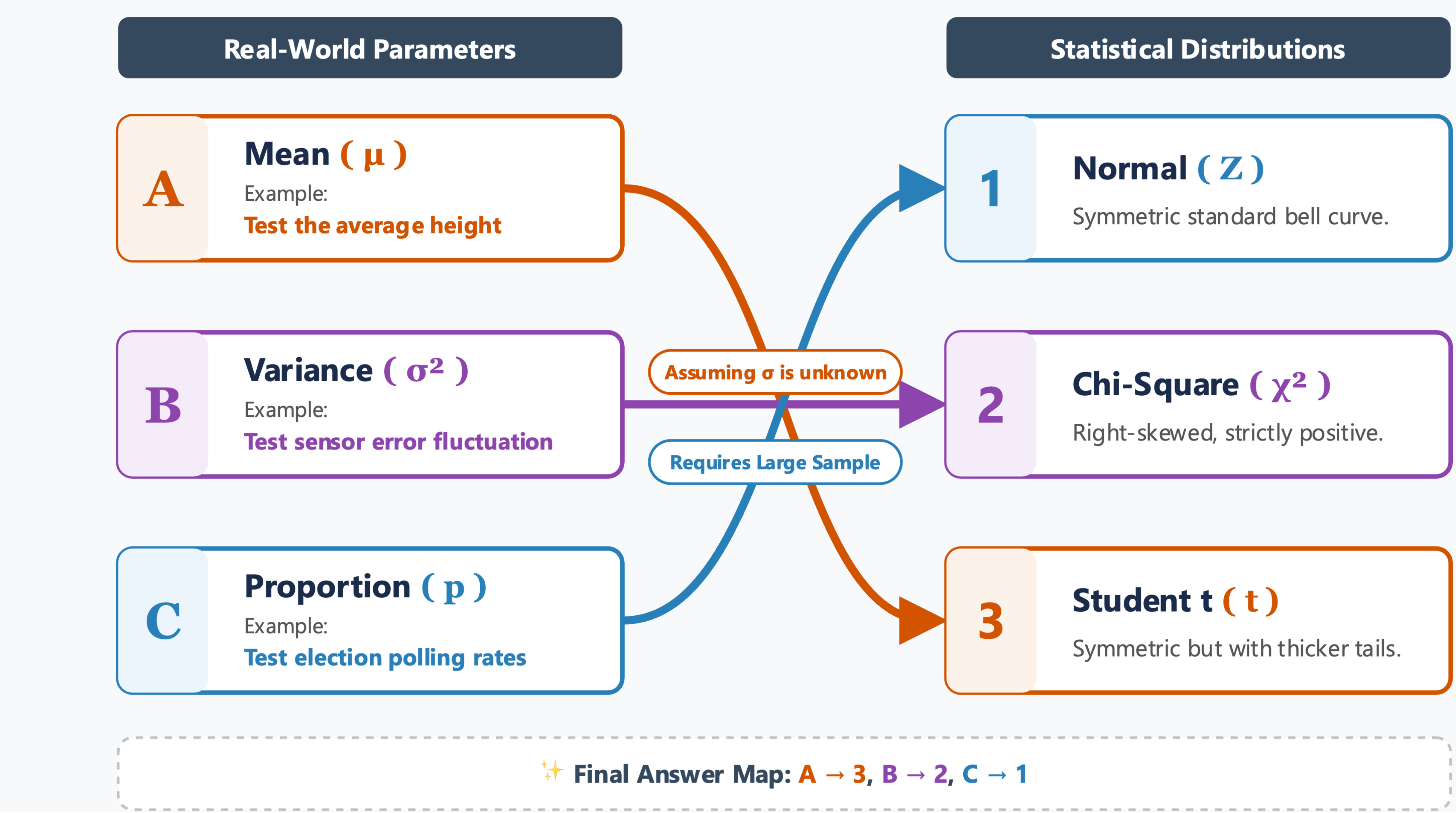
1. T
2. F → A larger α makes rejection of H_0 easier.
3. F → We can test means, proportions, variances, etc.

Q9. Parameter-to-Distribution Mapping (Matching)

Question

Parameters: A. Mean B. Variance C. Proportion

Distributions: 1. Normal 2. Chi-square 3. Student t


Real-World Parameters
Statistical Distributions
A
Mean (μ)
Example:
Test the average height
B
Variance (σ²)
Example:
Test sensor error fluctuation
C
Proportion (p)
Example:
Test election polling rates
Assuming σ is unknown
Requires Large Sample
1
Normal (Z)
Symmetric standard bell curve.
2
Chi-Square (χ²)
Right-skewed, strictly positive.
3
Student t (t)
Symmetric but with thicker tails.
✨ Final Answer Map: A → 3, B → 2, C → 1

Q10. Test Type, Direction, and Critical Region (Structured Multiple Choice)

For each alternative hypothesis, choose the correct test type.

1. $H_a : \mu \neq \mu_0$

   A. Left-tailed

   B. Right-tailed

   C. Two-tailed

2. $H_a : \mu < \mu_0$

   A. Left-tailed

   B. Right-tailed

   C. Two-tailed

3. $H_a : \mu > \mu_0$

   A. Left-tailed

   B. Right-tailed

   C. Two-tailed

4. The lecture states that different test types lead to different selected ____________ values.

   A. sample means

   B. z scores (critical values)

   C. variances

   D. confidence levels

Q10 — Test Type, Direction, and Critical Region

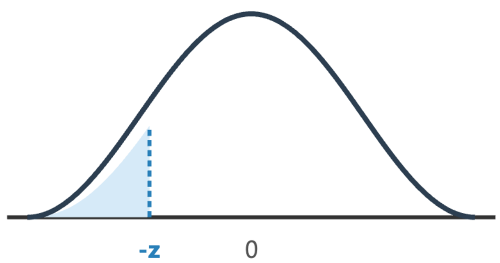
The symbol in the Alternative Hypothesis (H_a) acts as an "arrow" pointing to the rejection region.

<	$H_a: \mu < \mu_0$ **Left-Tailed Test**

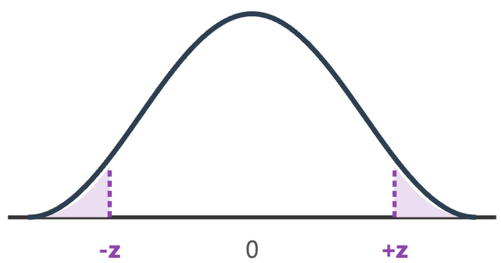
≠	$H_a: \mu \neq \mu_0$ **Two-Tailed Test**

>	$H_a: \mu > \mu_0$ **Right-Tailed Test**

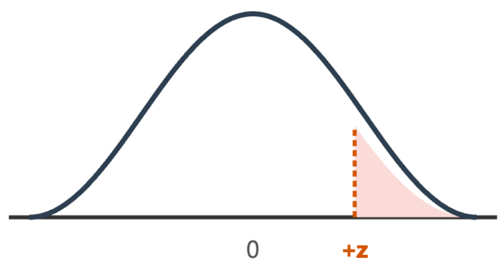
Rejection Region on the Left



Rejection Region split in Half



Rejection Region on the Right



Example: if $\alpha = 0.05$

Entire 5% error is in the left tail.

`Critical z = -1.645`

Example: if $\alpha = 0.05$

Error split to 2.5% per tail.

`Critical z = ±1.96`

Example: if $\alpha = 0.05$

Entire 5% error is in the right tail.

`Critical z = +1.645`

Question Answer (Part 4) & Ultimate Takeaway:

Because the rejection region shifts and splits, *Different test* **CRITICAL Z SCORES**

1. $H_a : \mu \neq \mu_0$ **C.** Two-tailed
2. $H_a : \mu < \mu_0$ **A.** Left-tailed
3. $H_a : \mu > \mu_0$ **B.** Right-tailed
4. The lecture states that different test types lead to different selected **critical** values.
   B. z scores (critical values)

Q11. Decision Rule and Conclusion Wording (Applied Reasoning)

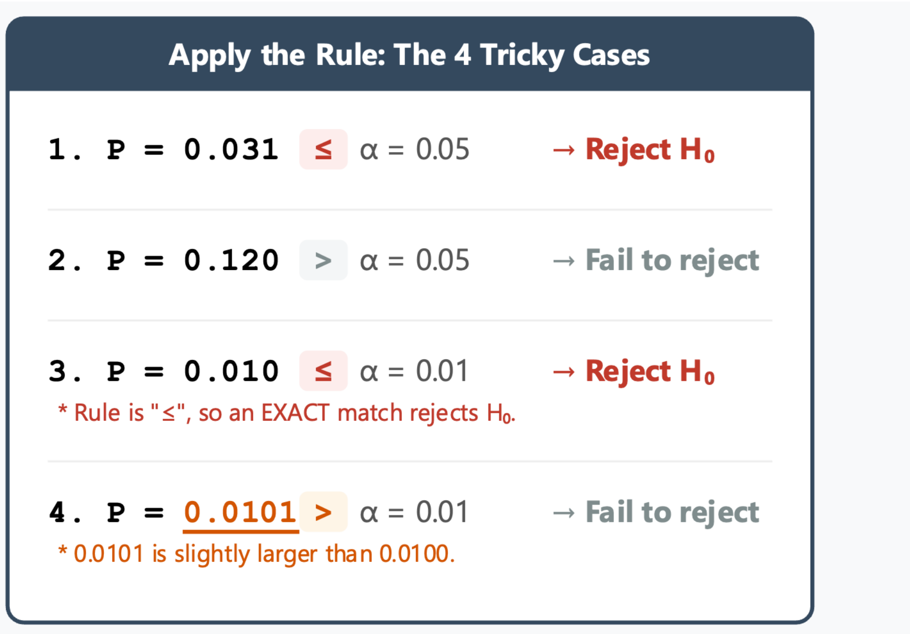
Use the lecture decision rule: If P-value $\leq \alpha$, reject H_0.

For each case, decide whether to Reject or Fail to reject H_0.

1. $P = 0.031, \ \alpha = 0.05$
2. $P = 0.120, \ \alpha = 0.05$
3. $P = 0.010, \ \alpha = 0.01$
4. $P = 0.0101, \ \alpha = 0.01$

Then complete the standard reporting phrase (one line):

"If we fail to reject H_0, there is ________________ evidence to support / warrant rejection of the claim."



1. **Case 1:** P=0.031, α=0.05. 0.031 is strictly less than 0.05. Because P is low, we **Reject H0**.
2. **Case 2:** 0.120 is clearly larger than 0.050. We do not have enough evidence, so we **Fail to reject H0**.
3. **Case 3:** Look at the rule carefully—it says 'less than OR EQUAL TO'. Since it's an exact tie, the rule is met. We **Reject H0**.
4. **Case 4:** 0.0101 is just a tiny fraction *larger* than 0.01. It crossed the line! Therefore, we **Fail to reject H0**.

Finally, for the conclusion wording: When we fail to reject the null hypothesis, we never say we 'accept' it. The legally correct statistical phrase is: 'There is **not sufficient** evidence to support/warrant rejection of the claim.

Q11. Decision Rule and Conclusion Wording (Applied Reasoning)

Use the rule: If P-value $\leq \alpha$, reject H_0.

1. $P = 0.031,\ \alpha = 0.05$ **Reject** H_0
2. $P = 0.120,\ \alpha = 0.05$ **Fail to reject** H_0
3. $P = 0.010,\ \alpha = 0.01$ **Reject** H_0
4. $P = 0.0101,\ \alpha = 0.01$ **Fail to reject** H_0

If we fail to reject H_0, there is **not sufficient** evidence to support / warrant rejection of the claim.

Q12. Two Samples, ANOVA, and Non-parametric Tests (Integrated Question)

Part A — Two-sample proportion hypotheses (Fill in the blanks)

Complete the equivalent null and alternative forms shown in the lecture:

$$H_0 : p_1 = p_2 \iff H_0 : p_1 - p_2 = \underline{\hspace{2cm}}$$

$$H_1 : p_1 \neq p_2 \iff H_a : p_1 - p_2 \neq \underline{\hspace{2cm}}$$

Part B — One-way ANOVA (Multiple Choice)

One-way ANOVA is used to test:

A. Equality of two population variances only

B. Equality of three or more population means

C. Whether a single sample mean equals a constant

D. Whether a population proportion equals 0.5

Part C — ANOVA requirements (Select all that apply)

Which conditions are listed in the lecture for one-way ANOVA?

A. Populations are approximately normal

B. Populations have the same variance (or standard deviation)

C. Samples are simple random samples of quantitative data

D. Samples are independent

E. Variables must be categorical only

F. Different samples are from populations categorized in only one way

Part D — Non-parametric tests (Short Answer)

State one advantage and one disadvantage of non-parametric tests given in the lecture.

Q12 — Advanced Testing Methods: ANOVA & Beyond

A comprehensive breakdown of Two-Sample, ANOVA, and Non-Parametric methodologies

Part A: Two-Sample Proportions

Algebraic Equivalence in Hypotheses

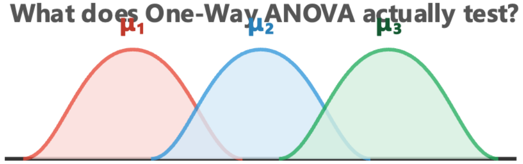
$p1 = p2 \rightarrow p1 - p2 = 0$

Why do we write it this way?

Stating things are equal is the same as saying the **Difference is Zero (0)**. This is standard for H_0.

Part B: The Purpose of ANOVA

What does One-Way ANOVA actually test?



THE BIGGEST TRAP:

Despite the name "Analysis of Variance", it is used to compare 3 or more **MEANS (μ)**!

Part C: ANOVA Requirements

Checklist of strict assumptions:

- ✓ A. Normal populations
- ✓ B. Equal/Same variances
- ✓ C. Quantitative data (Response Var)
- ✓ D. Independent samples
- ✗ ~~E. Variables must be categorical only~~
  (Groups are categorical, but data is numeric)
- ✓ F. Categorized in one way ("One-Way")

Part D: Non-Parametric Tests

If data fails ANOVA assumptions, we switch to Non-Parametric tests.

ADVANTAGE

"Distribution-Free" — No Normal distribution required! Great for outliers & ordinal data.

DISADVANTAGE

Less "Statistical Power". More likely to fail to reject a false H_0 (i.e., higher Type II Error).

Part A — Two-sample proportion hypotheses

$$H_0 : p_1 = p_2 \iff H_0 : p_1 - p_2 = 0$$
$$H_1 : p_1 \neq p_2 \iff H_a : p_1 - p_2 \neq 0$$

Part B — One-way ANOVA

B. Equality of three or more population means

Part C — ANOVA requirements

The conditions listed in the lecture are:

A. **A.** Populations are approximately normal

B. **B.** Populations have the same variance (or standard deviation)

C. **C.** Samples are simple random samples of quantitative data

D. **D.** Samples are independent

E. **F.** Different samples are from populations categorized in only one way

Part D — Non-parametric tests

One advantage:

- They have less rigid requirements and can be applied to a wider variety of situations.

One disadvantage:

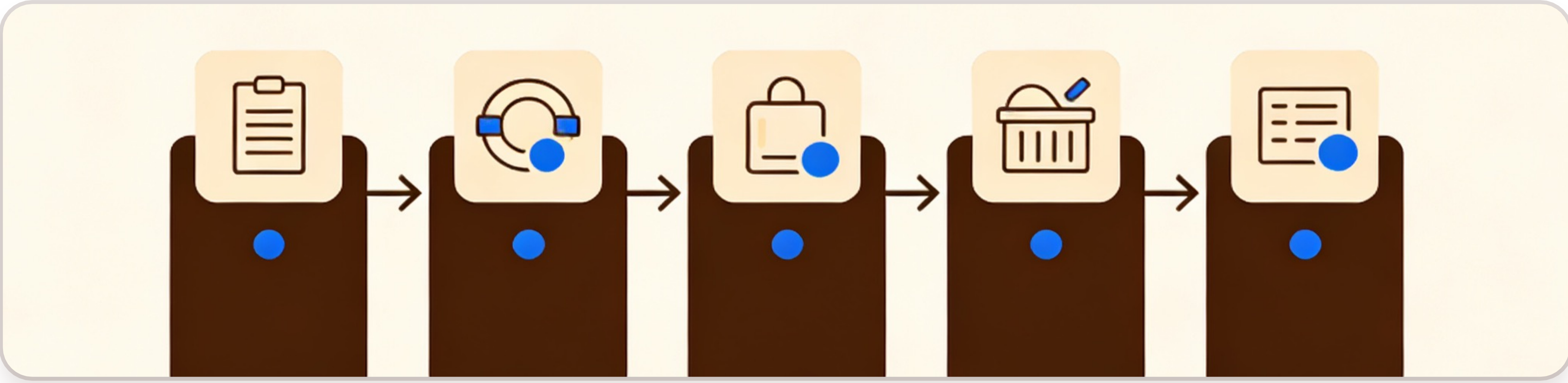
- They are not as efficient as parametric tests, so stronger evidence is generally needed to reject H_0.

ypothesis Testing（expand knowledge

A systematic procedure to evaluate claims about a population parameter.

STATISTICAL INFERENCE | DATA ANALYSIS | RESEARCH METHODOLOGY

The 5-Step Process of Hypothesis Testing



1. State Hypotheses

Define Null (H_0) and Alternative (H_1) clearly.

2. Set Significance

Choose α level (e.g., 0.05) for Type I error risk.

3. Test Statistic

Compute the statistic (e.g., t-value) from sample data.

4. Determine P-value

Calculate the probability of observing the data if H_0 is true.

5. Make a Decision

Compare p-value with α and decide to reject H_0 or not.

Case Study: Quality Control of a Coin Production Line

Scenario: A coin factory samples 20 coins to verify if the mean weight deviates from the standard 1.0g.

Step 1: State Hypotheses

H_0: μ=1.0g | H_1: $\mu \neq$1.0g (Two-tailed)

Step 2: Significance Level

Set α = 0.05 (Standard threshold)

Step 3: Test Statistic

Data: n=20, $\bar{x}$=0.96g, s=0.10g → t ≈ -1.79

Step 4: P-value Result

df=19 → P-value ≈ 0.09 (Not significant)

CONCLUSION: Since p-value (0.09) > α (0.05), we fail to reject the null hypothesis.

Decision and Conclusion

"The scale remains balanced. We find no significant difference to reject the standard weight."

Step 5: Make a Decision

P-value (0.09) > α (0.05) → **Fail to reject H_0**

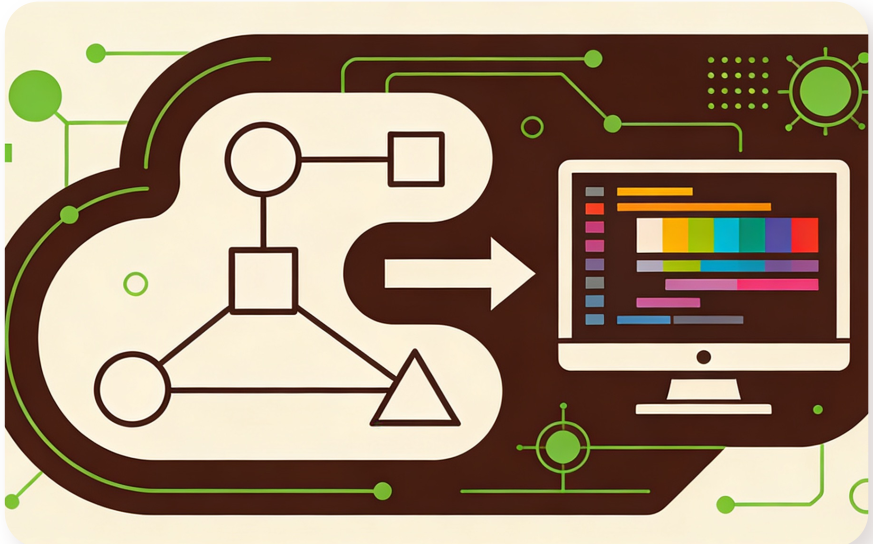
Key Conclusion

Insufficient evidence to say mean ≠ 1.0g. Observed difference is likely random.

P-value Meaning (0.09)

9% probability of such extreme results if H_0 is true — not rare enough.

How to Calculate t-value and P-value?



Calculating the t-value

Formula: $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$
Example: $t \approx -1.79$ (for sample n=20)

Calculating the P-value

Use software (Excel, Python, R) for precision.
Excel: =T.DIST.2T(ABS(-1.79), 19)
Result: **P-value ≈ 0.09**

Key Insight: Always use statistical software for precise P-value results to avoid estimation errors.